## CHAPTER QUESTIONS

SKILLS REASONING 7

1 Explain why lithium, sodium and potassium are put in the same group in the Periodic Table.

SKILLS CRITICAL THINKING

2 This question concerns the chemistry of the elements Li, Na, K, Rb and Cs. In a and b, you should name the substances represented by letters.

a **A** is the least dense of all the metals.

b When metal **B** is dropped onto water it melts into a small ball and moves rapidly around the surface. A gas, **C**, is given off and this burns with a lilac flame. A solution of **D** is formed.

9 c Write an equation for the reaction of **B** with water.

SKILLS REASONING 6 d What would you expect to see if solution **D** was tested with universal indicator paper?

7 e Explain why **B** melts into a small ball when it is dropped onto water.

SKILLS CRITICAL THINKING 9 f **E** burns in air with a yellow flame to form compound **F**. Write a word equation and balanced symbol equation for the reaction that occurs.

7 3 Explain whether each of the following statements is true or false.

a Sodium forms mostly covalent compounds.

8 b A rubidium atom is larger than a potassium atom.

6 c All the alkali metals react with air to form oxides.

7 d Lithium reacts with chlorine to form lithium chloride, which has the formula $Li_2Cl$.

SKILLS REASONING 8 4 Imagine that a new alkali metal has recently been discovered and that it fits into the Periodic Table below francium. We will call this new element edexcelium.

a Explain whether you would expect edexcelium to be more or less dense than francium.

SKILLS PROBLEM SOLVING 9 b State how many electrons edexcelium will have in its outer shell.

SKILLS CRITICAL THINKING 8 c State the names of the products that will be formed when edexcelium reacts with water.

d Explain whether edexcelium will be more or less reactive than francium.

SKILLS PROBLEM SOLVING 9 e If the symbol for edexcelium is Ed, write a balanced chemical equation for the reaction of edexcelium with water.

SKILLS CRITICAL THINKING 6 f When edexcelium reacts with water, will the solution formed be acidic, alkaline or neutral?

SKILLS PROBLEM SOLVING 7 g Write the formula for the compound formed when edexcelium reacts with air.